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GPU vs. CPU: A Comparative Study of Performance, Architecture, and NVIDIA's Innovations in Modern Computing

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***Abstract*—Graphics Processing Units (GPUs) and Central Processing Units (CPUs) are the foundations of contemporary computing in the rapidly developing field of computational technology. Their performance, architectural variations, and energy efficiency are compared in this research, emphasizing their distinct advantages across a range of workloads, including gaming, scientific computing, and artificial intelligence. This survey paper highlights how GPUs have transformed parallel processing with breakthroughs like Compute Unified Device Architecture (CUDA) cores and tensor operations. CPUs continue to play a crucial role in sequential and hybrid computing applications. Particular attention is given to NVIDIA's achievements, which include innovations in Turing and Ampere architectures, NVLink interconnects, and hybrid CPU-GPU systems such as the Grace Hopper superchip. The results highlight the complementary functions of these processors in heterogeneous systems and address their consequences for energy efficiency, High-Performance Computing (HPC), and hybrid architectures in the future.**

***Index Terms*—CPU, GPU, NVIDIA, Performance Comparison, Deep Learning, Hybrid Computing, Architecture, Parallelism.**

I. Introduction

The rise of Artificial Intelligence (AI) and High Performance Computing (HPC) has led to a shift in modern computational systems with traditional (Central Processing Units) CPUs being replaced by GPUs (Graphics Processing Units). GPUs are now at the forefront of deep learning, big data analytics, and molecular modeling. This transition is supported by innovations in hybrid CPU-GPU systems, led by NVIDIA which has developed groundbreaking GPU technology strategies like CUDA, NVLink, and tensor cores. GPUs are seen as essential enablers in AI and HPC advancements, while CPUs are adapting with more cores, better instruction sets, and energy-efficient designs to keep up with the market led by GPUs.

A. Motivation and Importance of GPUs and CPUs in Computing

Modern systems have evolved due to advancements in computational hardware. CPUs originally designed for sequential operations have adopted multi-core designs and updated instruction sets, while GPUs originally for graphic rendering are well-suited for parallel processing [1], [2].

B. Overview of GPUs and CPUs in High-Performance Computing (HPC)

HPC is crucial for scientific research, data-intensive applications, and AI. CPUs and GPUs are essential components, each offering unique advantages. CPUs manage intricate sequential tasks and handle various workloads due to their intricate memory structures [3] and [1] GPUs significantly improve HPC performance standards by speeding up tasks like real-time data analytics and chemical simulations with NVIDIA's NVLink interconnect, enhancing high-bandwidth, low-latency communication between GPUs and CPUs [4], [5].

C. Scope and Objectives

Furthermore, this study aims to provide a comprehensive analysis of the architectural and performance differences between CPUs and GPUs, as well as their complementary roles in modern computing. This research evaluates the benefits and drawbacks of both processors under various workload conditions. The objectives of this study include the following:

- **Analyzing Architectural Differences:** analyzing how CPUs and GPUs differ in terms of their processing cores, memory architectures, and data processing capacities.
- **Performance Comparison:** energy efficiency, thermal performance, and computational speed are examples of workload-specific performance measures that are evaluated using evidence from [6].
- **Highlighting NVIDIA's Innovations:** investigating cutting-edge technologies that have established GPUs as leaders in HPC, such as CUDA, tensor cores, and NVLink, as mentioned in [7].
- **Future Directions in Hybrid Systems:** evaluating CPU-GPU hybrid architectures' ability to alleviate memory and data transmission constraints [8].

II. Related Works

Studies on CPUs and GPUs in computing reveal their differences in sequential and parallel processing. GPUs excel in data-parallel applications like deep learning and scientific simulations, while CPUs excel in tasks requiring intricate sequential logic. The work in [2] and [3] provides a comprehensive understanding. The study [9] emphasizes the importance of power management in GPU-centric systems, while [10]

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explores the potential of programming models like OpenCL and CUDA to enhance CPU-GPU cooperation. Research shows architectural developments have transformed GPU capabilities with NVIDIA's innovations like NVLink addressing data transfer inefficiencies between GPUs and CPUs in hybrid system research [5], [9].

A. Gaps in Existing Research

Even though the field of CPU-GPU comparison research has expanded significantly, there are still several important gaps that prevent a thorough grasp of how to best utilize these processors for contemporary computational tasks. The majority of current research concentrates on certain features of CPU or GPU architectures, performance metrics, and power usage, but it does not offer a cohesive perspective of their comparative advantages in many settings. The key gaps identified in existing literature include limited exploration of hybrid architectures, lack of detailed comparative study between CPUs, GPUs, and other accelerators, inadequate focus on cross-platform optimization, absence of comprehensive benchmarking for hybrid systems, energy efficiency in hybrid architectures and scaling to exascale and beyond. This survey paper tries to contribute to these sectors and in the following sections, these segments will be covered.

B. Contribution of This Survey

Even while research on CPUs, GPUs, and their cooperative designs has advanced significantly, there are still numerous gaps and restrictions in the surveys that are now available. Workload partitioning, programming models, and predictive modeling techniques are only a few of the specific topics covered in earlier studies in [2], [11]. The same is true for energy efficiency surveys, such as [9] which provide useful information about GPU optimization but do not provide a more comprehensive comparison of hybrid systems and new developments in hardware.

This survey addresses these gaps by providing:

- A Unified Comparison Across Performance, Architecture, and Innovations
- In-Depth Exploration of NVIDIA's Role in GPU Innovations
- Emphasis on Emerging Hybrid Architectures
- Forward-Looking Analysis of Future Trends:

III. CPU and GPU Architectures

The development of CPUs and GPUs has revolutionized computing, enabling advancements in data-intensive applications, high-performance computing, and AI. GPUs are specialized hardware for rendering graphics, while CPUs serve as the general-purpose computational foundation. This section compares their architectures.

A. CPU Architecture Overview

CPUs are made to be adaptable, they can efficiently perform a wide range of computational activities. They have been improved with multi-core designs to increase performance on parallel applications. CPUs are crucial for tasks involving intricate data relationships and decision-making.

1) Sequential Processing and Multi-Core Designs: Sequential processing in CPU architecture optimizes complex branching tasks, ensuring efficiency in multi-threaded applications and operating systems, thereby enhancing the performance of CPUs [2].

CPU architecture is transitioning from single-core to multi-core designs, enabling concurrent operation of multiple threads, especially useful for scientific simulations and virtualization, maximizing resource utilization [11]

2) Cache Memory and Latency Optimization: CPUs use hierarchical caches like L1, L2, and L3 to store frequently accessed data, improving speed and reducing latency. Advanced prefetching methods anticipate data usage for subsequent operations [3].

CPUs optimize instruction pipelines using branch prediction units, reducing pipeline stalls and enabling faster processing making them highly effective in control systems and financial modeling applications requiring precision and latency [12].

B. GPU Architecture Overview

GPUs are designed for workloads requiring massive parallelism, excelling in real-time rendering, deep learning, and numerical simulations with their architecture prioritizing performance for efficient dataset management.

1) Parallelism and Massive Multi-Threading: GPU architecture's parallelism allows for efficient job breakdown into smaller calculations with lightweight cores arranged into Streaming Multiprocessors (SMs), resulting in great processing performance [2]. GPUs use massive multi-threading for scalability, unlike CPUs which prioritize single-threaded speed. SMs manage thousands of concurrent threads, enabling better performance in data-parallel jobs like matrix multiplications, where the same operation is performed on large datasets [12].

2) CUDA Cores and Tensor Cores in Modern GPUs: GPUs, powered by NVIDIA's CUDA architecture are powerful general-purpose processors capable of high-performance computing applications. CUDA cores, lightweight processing units, perform tasks in parallel. Tensor cores, introduced in NVIDIA's Volta architecture and improved in Ampere GPUs, speed up matrix computations, enabling mixed-precision calculations for neural network training and inference [7].

GPUs with CUDA and tensor cores, enhance speed making them crucial in tasks like scientific modeling, deep learning, real-time analytics, and medical imaging.

C. Comparison of Architectural Designs

Modern computing relies heavily on CPUs and GPUs with GPUs designed for high-throughput parallel processing and CPUs for sequential, low-latency processing.

1) Number of Cores and Memory Structures: CPUs have limited cores (2-16), while GPUs like NVIDIA A100 have thousands, optimized for parallel tasks. GPUs use high-bandwidth memory for large data tasks, prioritizing throughput over latency [8].

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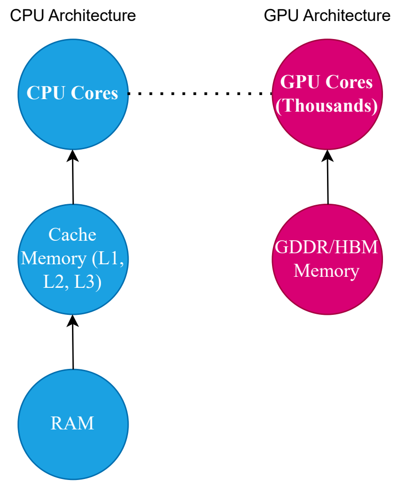

Fig. 1. Comparison of CPU and GPU architecture.

2) Processing Power and Latency trade-offs: CPUs are ideal for real-time applications, while GPUs prioritize throughput and can process more data in parallel making them ideal for data analytics and deep learning [3].

3) Applications suited for CPU vs GPU: CPUs are suitable for high precision, sequential processing, and low latency tasks like operating systems and database management, while GPUs are better suited for high parallelism tasks [4]. Table I summarizes the architectural differences between CPU and GPU.

TABLE I
COMPARISON OF CPU AND GPU ARCHITECTURES

Feature	CPU	GPU
Cores	2-64	Thousands
Memory	Cache (L1, L2, L3)	HBM, GDDR
Performance	Single-threaded	Multi-threaded
Applications	General-purpose	Parallel tasks

Figure 1 highlights the key architectural differences between a Central Processing Unit (CPU) and a Graphics Processing Unit (GPU), focusing on core design, memory structure, and data flow. CPUs are designed for sequential processing with powerful cores and cache memory making them ideal for tasks like operating systems and database management. GPUs, on the other hand, have thousands of smaller cores optimized for parallel processing, making them efficient for large datasets. Data flow between CPUs and GPUs is facilitated by RAM to cache and GDDR/HBM memory, allowing simultaneous computing. In a hybrid CPU-GPU system, GPUs speed up parallel operations while CPUs manage sequential tasks, a crucial aspect of AI and high-performance computing.

IV. Performance Comparison

This section compares CPU and GPU performance under different scenarios, focusing on their architectural designs with CPUs designed for low-latency, single-threaded applications and GPUs for multi-threaded activities.

A. Single-Threaded vs Multi-Threaded Performance

CPUs are ideal for sequential processing and low-latency tasks, such as high-performance computing, database queries, and operating system administration, due to their single-threaded performance [2].

GPUs outperform CPUs in deep learning, handling thousands of threads simultaneously and outperforming them 10x to 100x in real-time data processing [13].

B. Workload-specific Performance Comparison

This section discusses the performance of CPU and GPU in various fields such as matrix multiplication, graphics rendering, and deep learning.

1) Matrix Multiplications and Scientific Computations: Matrix multiplication is crucial in scientific computer operations like numerical analysis and simulations. CPU performance is limited by cores, but GPUs can calculate thousands of matrix elements simultaneously, enabling faster multiplications [3].

For example, the Ampere-based NVIDIA A100 GPU can perform matrix operations at up to 100 teraflops (TFLOPS) [7], but the top-tier CPUs, such as the Intel i9 chip can only perform the same operations at up to 3.5 gigaflops (GFLOPS). This significant disparity demonstrates the benefits of GPUs for repeated, high-volume mathematical processing applications.

2) Deep Learning Model Training and Inference: GPUs, like NVIDIA's Turing and Ampere architectures, are optimized for deep learning by incorporating Tensor Cores which accelerate matrix computations while maintaining accuracy [7]. GPUs excel in AI inference due to their ability to process large amounts of data quickly and handle massive parallelism, reducing training time from days to hours [14]. However, CPUs are still relevant for tasks involving smaller models or inference, where low latency and rapid response times are more critical than raw computational power [15].

3) Graphics Rendering and Gaming: GPUs are initially created for graphics rendering and even today it is one of its core functions. They are suitable for this job as they can perform the same operation on many pixels simultaneously. A key aspect of gaming, real-time rendering such as lighting, shadows, and textures, in a highly parallelized fashion performs a series of repetitive mathematical calculations. NVIDIA's Turing architecture features Ray Tracing Cores, enhancing real-time graphics rendering by computing reflections, lighting, and shadows, enabling GPUs like RTX 3080 to render scenes with unprecedented realism [16].

C. Energy Efficiency and Thermal Considerations

The need for heat control and superior efficiency has grown along with the demand for processing power. The benefits and drawbacks of CPU and GPU in terms of efficiency and heat management vary depending on the specific workload.

1) Power consumption in CPUs and GPUs: CPUs are more efficient and energy-efficient in low parallelism activities due to their reduced core count and power-saving features, while GPUs are more power-intensive due to their higher core count and high performance [9]. GPUs are often more power-efficient

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than CPUs, particularly in parallel activities like deep learning which is crucial in data centers due to high energy costs [17].

2) Thermal design and heat dissipation: Due to GPUs having higher performance, they generate more heat as well. Therefore, efficient thermal design is an integrated part of designing GPU systems. The significance of GPU thermal design optimization, especially in high-performance computing settings where prolonged workloads may cause thermal throttling if cooling is inadequate. Modern GPUs, like NVIDIA RTX and Ampere series utilize multiple fan systems, heat pipes, and vapor chambers to effectively dissipate heat generated by parallel cores. CPUs require less cooling due to fewer cores and electricity usage, but liquid cooling solutions become increasingly important in high-performance workstations and servers [18].

Table II shows the differences between CPUs and GPUs in various workloads.

TABLE II
PERFORMANCE COMPARISON ACROSS WORKLOADS

Workload	CPU Performance	GPU Performance	Speedup Factor
Matrix Multiplication	3.5 GFLOPS	100 TFLOPS	28500x
AI Inference (Latency)	500 ms	5 ms	100x
Gaming (FPS)	60 FPS (Intel i9)	120 FPS (RTX 3080)	2x
Scientific Simulation	Days to Weeks	Hours to Days	10-100x

V. NVIDIA's Innovations in GPU Technology

NVIDIA has been a leader in GPU technology, improving graphic rendering and transforming high-performance computing and AI. Their architectures, including Turing, Ampere, and Grace Hopper have advanced real-time ray tracing, AI optimization, and hybrid CPU-GPU integration, enabling them to dominate AI, HPC, and gaming industries.

A. NVIDIA's NVLink

A high-bandwidth, low-latency interconnect called NVIDIA's NVLink improves communication between GPUs in a system with several GPUs. With a maximum bandwidth of 25 Gbps—a substantial improvement over the 16 Gbps maximum of Peripheral Component Interconnect Express (PCIe), NVLink significantly outperforms the conventional PCIe interconnect. Overcoming the CPU and minimizing communication bottlenecks, NVLink's architecture allows GPUs to communicate directly with one another, as explained in [5]. Large systems like NVIDIA's DGX-2, which combines eight GPUs into a single workstation have been significantly impacted by NVLink. This method enables more efficient collaboration among numerous GPUs. According to [5], this interface allows for amazing scalability which is crucial for expanding the possibilities of contemporary computer applications.

1) Tensor Cores and Ray Tracing Cores: Turing introduced Tensor Cores and could handle high-speed matrix multiplications required for deep learning tasks. Tensor Cores help GPUs to take care of mixed-precision computations which accelerate AI model training and inference without a trade-off in accuracy. For deep learning tasks, this capability is very beneficial as there is a need to perform repeated matrix operations by large neural networks. Ray Tracing (RT) Cores in Turing architecture enable real-time ray tracing on consumer GPUs, enhancing visual fidelity in simulations and video games by showcasing realistic lighting and reflections [16].

2) Performance Impact on Gaming and AI: Turing architecture's performance impact in gaming and AI industries has been recognized with real-time ray tracing enhancing in-game landscapes and Tensor Cores enabling significant speedups for AI in deep learning inference and model training. Games like Cyberpunk 2077 and Shadow of the Tomb Raider showcase these capabilities. Turing GPUs are increasingly popular for gaming and AI researchers [16].

B. Ampere Architecture: Scaling AI and HPC Workloads

Expanding on Turing's accomplishments, NVIDIA's Ampere architecture which was revealed in 2020, increased the GPUs' potential for AI and HPC workloads. Ampere's inclusion of next-generation Tensor Cores and Multi-Instance GPU (MIG) technology allowed for more flexibility and effectiveness in handling a variety of tasks [7].

1) Mixed-Precision Operations and Tensor Cores: The Ampere design supported TF32 (TensorFloat32) precision which improved the mixed-precision capabilities of Tensor Cores. TF32 speeds up matrix computations while maintaining accuracy levels suitable for AI model training. Large neural network training times were greatly reduced by this invention, allowing for quicker iterations in AI research and development [7].

2) Improvements in AI Inference and Model Training: Ampere architecture enhances AI inference and model training by supporting Multi-Instance GPU (MIG) technology, allowing simultaneous execution of multiple workloads on a single GPU [7]. Ampere GPUs like NVIDIA A100 are ideal for scientists developing parallel deep learning models, offering over five times higher training performance for applications like natural language processing, autonomous driving, and medical imaging [7].

C. Grace Hopper: Hybrid CPU-GPU Architectures

NVIDIA's Grace Hopper architecture, introduced in 2022, integrates CPU and GPU processing for AI and HPC workloads, addressing memory bandwidth, latency, and power efficiency challenges. Figure 2 represents the evolution of NVIDIA's GPU architectures over time.

D. Integration of CPU and GPU for AI and HPC

The Grace Hopper architecture which combines the Grace CPU with NVIDIA's Hopper GPU, optimizes CPU and GPU benefits in a single device. This design accelerates data transfer, making it ideal for AI training and HPC simulations, ensuring better load balancing and task handling in AI workloads [18].

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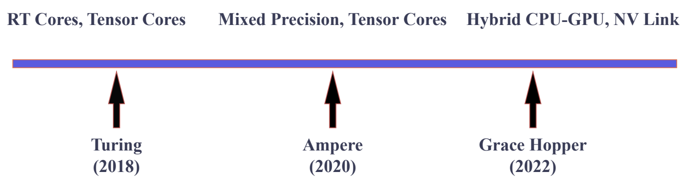

Fig. 2. NVIDIA architecture timeline: Turing to Grace Hopper.

VI. Hybrid CPU-GPU Systems

Hybrid CPU-GPU systems combine CPU's strengths for sequential tasks and GPU's efficiency for large datasets, enabling complex tasks like AI, scientific simulations, and data analytics with improved efficiency and speed.

A. The Need for Hybrid Systems in Modern Computing

When working with massive datasets or highly parallel computations, CPU performance reaches an equilibrium, despite their flexible task management and low latency responses [2]. Hybrid CPU-GPU systems combine CPUs for sequential logic and decision-making, while GPUs handle computationally demanding tasks like matrix multiplication in AI model training. Efficiency and scalability are especially crucial in the context of high-performance computing (HPC) [19].

B. Advantages of CPU-GPU Hybrid Systems

Comparing hybrid CPU-GPU systems to traditional single-processor systems reveals a number of advantages. Applications that demand both precise control over task execution and heavy data processing clearly benefit from these features.

1) Accelerating AI Workloads: It has been shown that hybrid systems are essential for speeding up AI tasks. In neural network training, GPUs handle the heavy lifting due to their immense parallelism, while CPUs handle scheduling and data distribution. Direct GPU-to-GPU communication is made possible by technologies like NVIDIA's GPUDirect, which minimizes latency and avoids the CPU [20]. Optimizing CPU-GPU workload distribution in GPU-accelerated data centers reduces operating costs and training times, crucial for AI applications like natural language processing and driverless cars [21].

2) Optimizing Scientific Simulations: Hybrid systems in scientific simulations balance logical processing with computational demands, like molecular dynamics, using GPUs for numerical calculations and CPUs for data collection and logic management [22]. Hybrid architectures, like NVIDIA DGX systems, are ideal for large-scale simulations like genomics and climate modeling due to their complementary CPU and GPU capabilities [5].

C. Challenges of Hybrid Architectures

Hybrid CPU-GPU systems offer significant performance boosts while they also present unique challenges. These difficulties are mostly related to memory hierarchy problems and data transfer bottlenecks that occur when using both processor types simultaneously.

1) Data transfer bottlenecks: In hybrid systems, data transfer between CPUs and GPUs continues to be a major bottleneck. PCIe and other traditional interconnects have trouble offering the latency and bandwidth needed for smooth connection in data-intensive tasks. This restriction often leads to inefficient performance and underuse of GPU resources [5].

NVLink and NVSwitch technologies, which enable high-bandwidth communication between CPUs and GPUs, have mitigated issues but scaling them across distant systems remains a challenge [5].

2) Memory hierarchy and latency issues: Complexity is increased by differences in memory hierarchies between CPUs and GPUs. CPUs employ hierarchical caches to reduce latency, while GPUs rely on high-bandwidth memory (HBM) that is optimized for huge datasets. Programming and resource management becomes more challenging when these architectures are synchronized [19]. Furthermore, handling huge datasets is made difficult by the GPU's memory's smaller size than system memory which necessitates frequent data transfers that worsen latency problems [19].

Table III depicts the advantages and challenges of hybrid CPU-GPU systems, highlighting factors like energy efficiency, memory architecture, and data transfer bottlenecks.

TABLE III
Comparison of GPU advantages and challenges

Aspect	**Advantage**	**Challenge**
Energy Efficiency	Better performance per watt for parallel tasks	Higher power consumption than CPUs
Memory Architecture	High-bandwidth memory for large data tasks	Latency in transferring data between CPU and GPU
Data Transfer	Faster execution using both processors	Data transfer bottlenecks between CPU and GPU

VII. Future Trend: Challenges and Limitations

This section examines the future of high-performance computing (HPC) by examining issues and developments in CPU-GPU hybrid systems, focusing on power consumption, memory integration, cost, and software compatibility.

A. Future Trends in CPU-GPU Hybrid Systems

Advanced interconnects, like NVLink-C2C and NVSwitch, enable quicker and more scalable connectivity between CPUs and GPUs. AI-specific accelerators, which offer specialized processing units for machine learning applications are expected to be integrated with conventional CPUs and GPUs in hybrid systems, further optimizing performance for AI workloads [18], [23].

B. Power and Thermal Challenges in GPUs

For GPUs, power consumption and heat generation are major issues, especially as processing power and core counts rise. High-end GPUs include integrated thermal management technologies, such as liquid cooling and sophisticated heat sinks,

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but these come at a cost and add complexity. Energy-efficient architectures must be given top priority in future designs to minimize environmental effects without sacrificing performance [24].

C. The Role of Memory in CPU-GPU Systems

In hybrid systems, memory architecture is still difficult. CPUs employ massive, low-latency caches for sequential workloads, while GPUs use high-bandwidth memory (HBM) for parallel processing. Latency and resource allocation problems arise when these dissimilar memory systems are integrated [19].

The goal of emerging solutions like on-package HBM for CPUs and unified memory is to close this gap. Particularly in exascale computing environments, these strategies facilitate smooth collaboration between CPUs and GPUs and lower data transportation overhead [14].

D. The Cost of High-Performance Hardware

Accessibility for researchers and smaller businesses is restricted by the expensive cost of high-performance CPUs, GPUs, and hybrid systems. Although GPUs, such as NVIDIA's A100, offer remarkable performance, their cost and power consumption restrict their applications. For wider usage, less expensive options and better designs will be essential [9].

E. Final Thoughts on The Future of CPU and GPU

CPUs and GPUs will continue to develop together, enhancing one another in increasingly integrated designs. CPUs will continue to be essential parts of hybrid systems, even though GPUs may eventually win in specialization and dominance in data-intensive industries. They will continue to collaborate, redefining computing capabilities and opening the door for advances in AI, HPC, and other fields.

VIII. Conclusion

This survey explores the roles of CPUs and GPUs in modern computing, highlighting their advantages and disadvantages. GPUs excel in parallelized workloads, while CPUs excel in sequential and control-intensive tasks. Hybrid systems, such as NVIDIA's NVLink, Tensor Cores, and unified memory architectures, address the increasing computing demands of AI, HPC, and scientific research. However, challenges like power consumption, memory integration, and software complexity need to be addressed. Future developments in CPU-GPU hybrid systems will be influenced by trends like scalable interconnects, AI-specific accelerators, and unified memory.

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